

IOR 4.0.0 HDF5 Optimization Session -- Quick Start

This package is a self-contained reproduction of session `ior/20260724_175545`: `an` `annotate` -> `baseline-trace` -> `diagnose` -> `4-dimension-optimize` pipeline run against IOR 4.0.0 (HDF5 1.14.5, Autotools) on Tuolumne.

Bottom line: no measured optimization beat the plain IOR baseline (`-a HDF5 -b 16m -t 4k -s 32 -C -F`, 512 ranks/8 nodes) outside its own measurement noise. Full detail, every number, every rejected variant, the session's chronological narrative, and a resume-point to-do list for a future session are in `REPORT.md` (also rendered as `REPORT.pdf`).

Reproduce it

1. Fill in `config.ini`: set `WORKSPACE_ROOT` to a scratch directory and `OUTPUT_ROOT` to a directory isolated from any real run data.
2. `bash scripts/install.sh` -- rebuilds HDF5 1.14.5 from source, installs `dftracer` (FUNCTION mode only, never `LD_PRELOAD`), builds the annotated IOR binary against both.
3. `bash scripts/run_all.sh <alloc-id>` -- runs the baseline and every optimization-variant script in order, each into its own fresh subdirectory under `$OUTPUT_ROOT`.
4. Compare the printed write/read median bandwidth against the ranges in `REPORT.md` Section 7b -- reproduction is successful if the new numbers fall within (or close to, accounting for site-specific filesystem drift) the reported ranges.

What's in here

Path	Contents
<code>REPORT.md</code> / <code>REPORT.pdf</code>	Full report: bottleneck diagnosis, every optimization variant tried (applied and rejected), the N-way comparator table, chronological session narrative, and a resume-point to-do list
<code>config.ini</code>	The ONLY file with real path/session values -- everything else is placeholder-only
<code>scripts/lib_load_config.sh</code>	Sourced by every other script; reads <code>config.ini</code>
<code>scripts/install.sh</code>	Rebuilds HDF5 + <code>dftracer</code> + annotated IOR
<code>scripts/run_*.sh</code>	One wrapper per run this session actually executed (baseline, I/O variants, communication variants, config/build helper scripts, incrementally discovered from every <code>tmp/* .sh</code> this session used)
<code>scripts/render_pdf.sh</code> + <code>scripts/_render_pdf.py</code>	Regenerates <code>REPORT.pdf</code> / <code>README.pdf</code> from the markdown (pure-Python, no <code>sudo</code> -- see script header)
<code>patches/annotated.patch</code>	Full source diff: <code>dftracer</code> annotation macros added to IOR's C sources, plus the <code>AC_PREREQ</code> build fix
<code>plan/</code>	The pipeline plan this session actually executed
<code>PERFORMANCE.md</code>	Pipeline self-profiling (step timings, retries)

Notes

- Every optimization-variant script launches with `dftracer` tracing **disabled** (`DFTRACER_ENABLE=0`) except the dedicated `baseline-trace` script -- this session found and fixed a real methodology bug where mixing traced and untraced arms in one A/B inflated apparent gains by ~60% (see `REPORT.md` Lessons Learned / Session Narrative).
- No script in this package hardcodes an absolute path, username, or allocation id -- everything routes through `config.ini` via `scripts/lib_load_config.sh`.
- Raw run/build logs are intentionally NOT included in this package (they stay in the session workspace's own `artifacts/` dir) -- only scripts, patches, params, config, and narrative docs are shipped here.